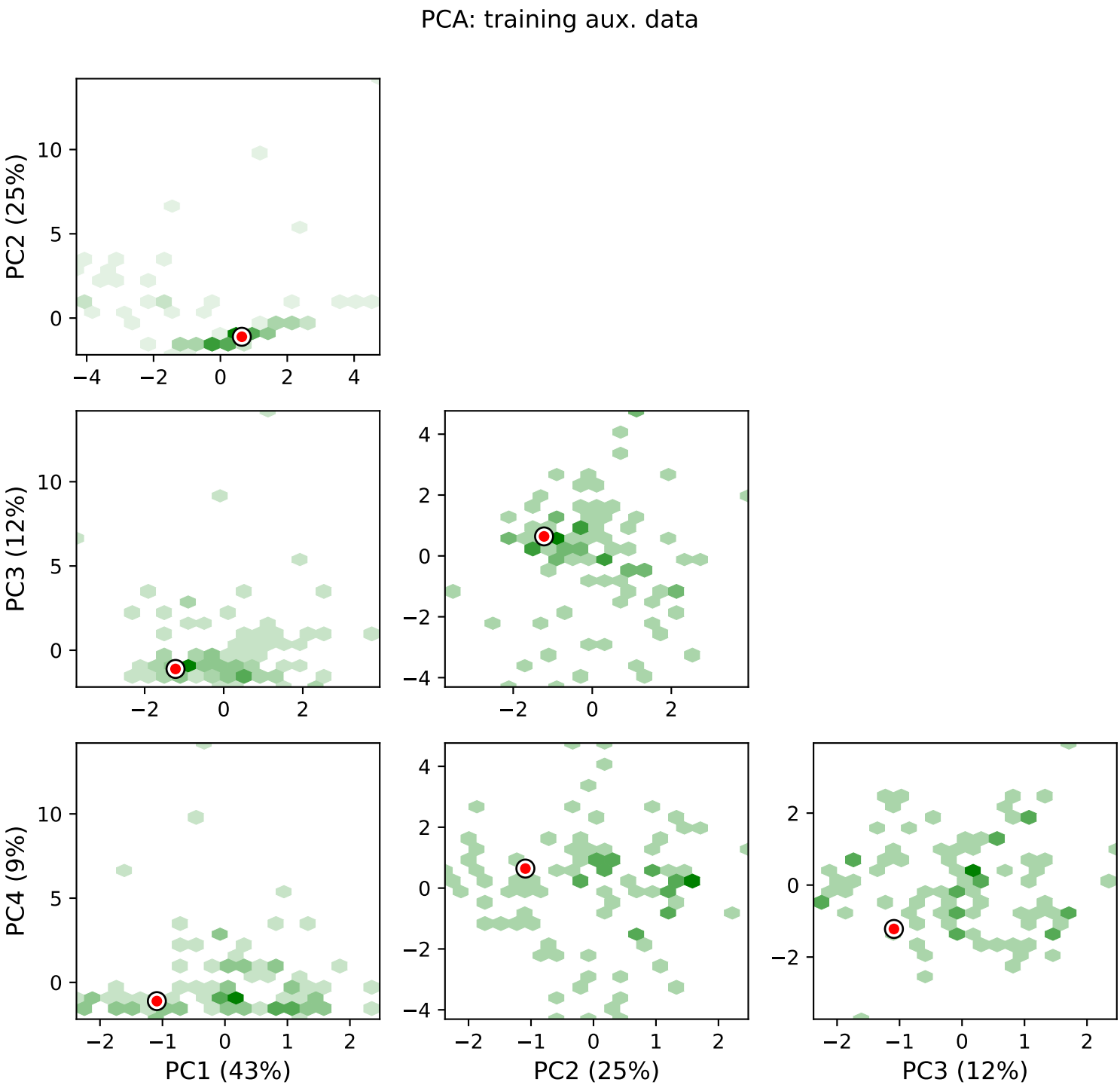
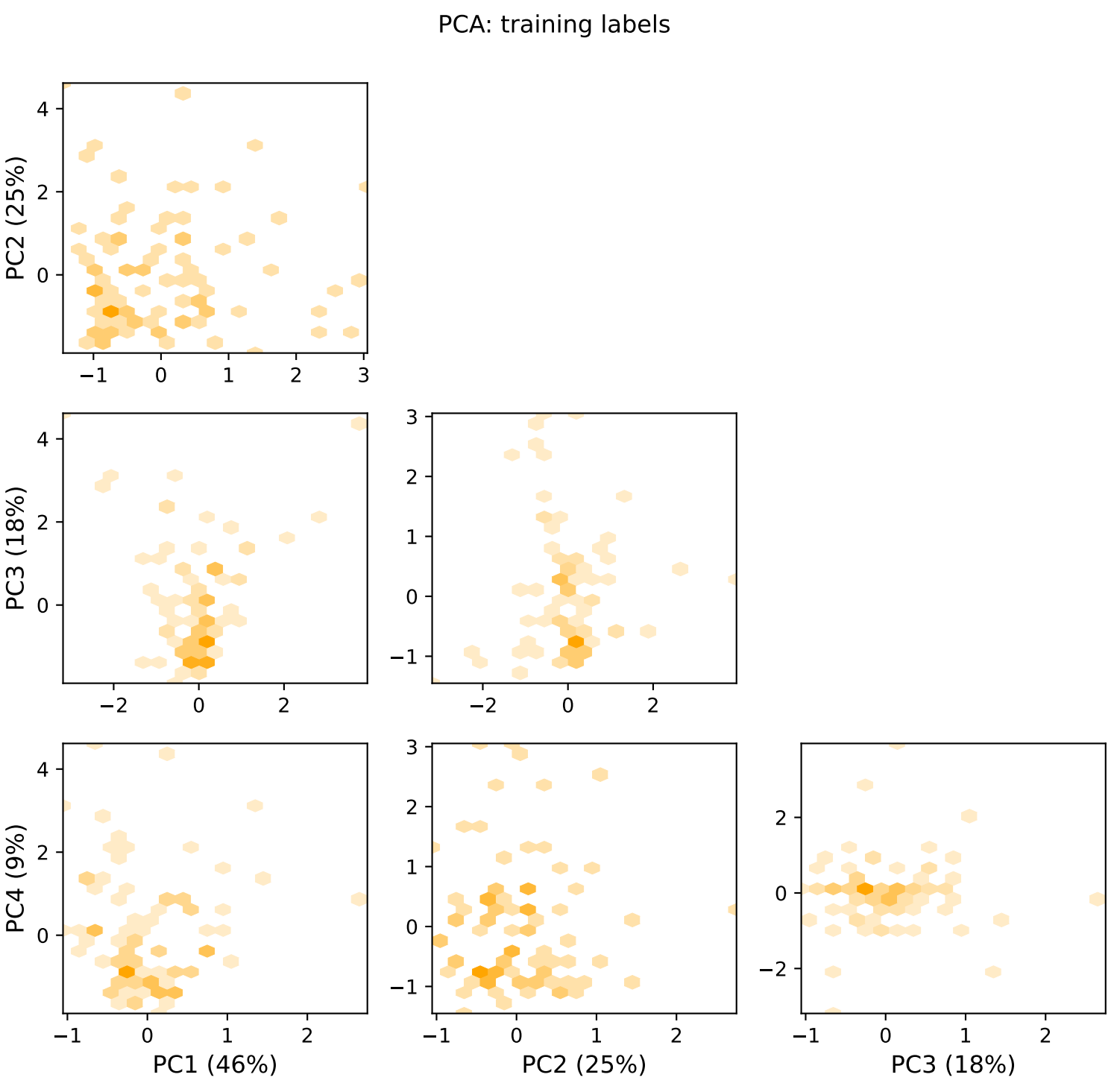
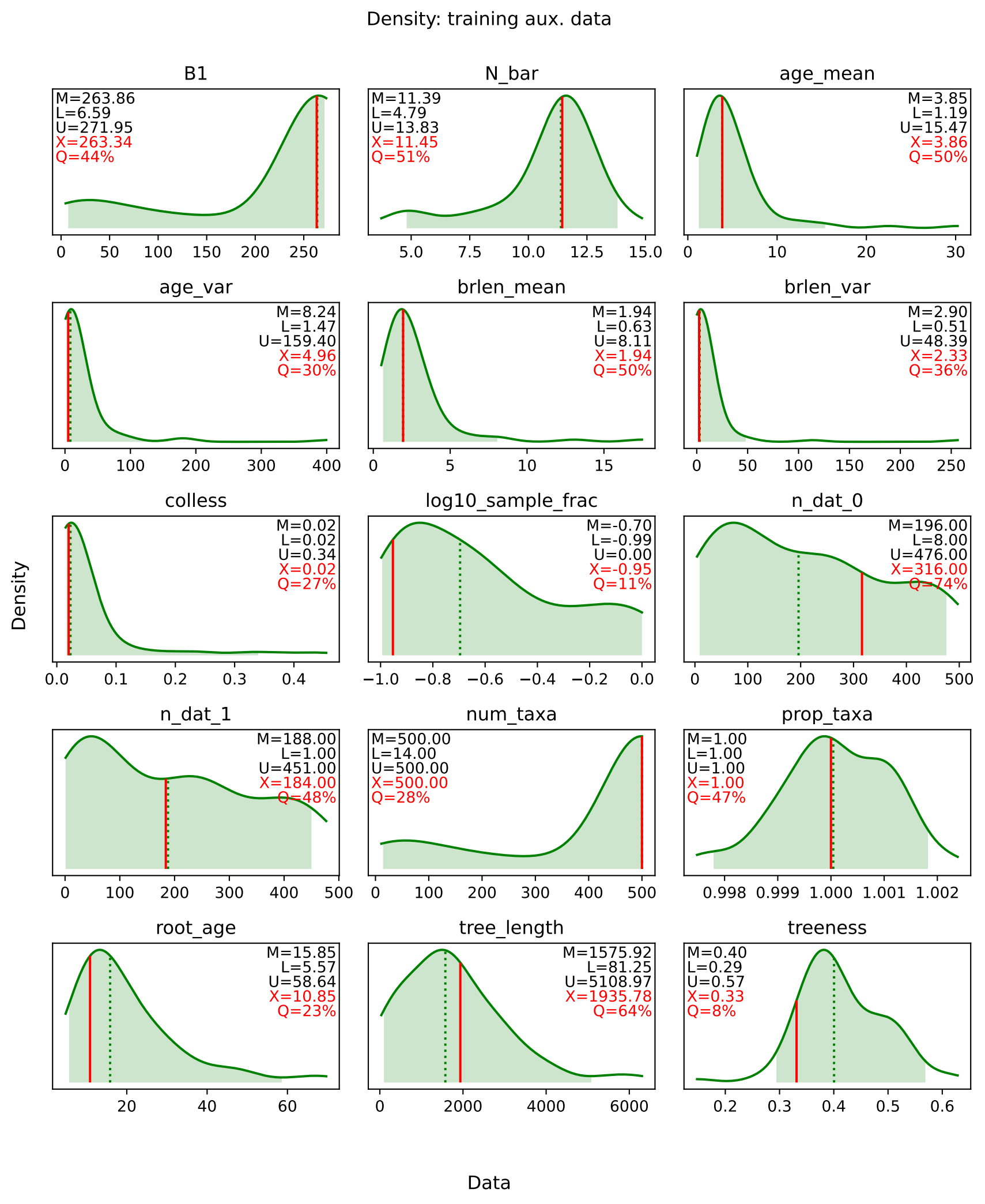
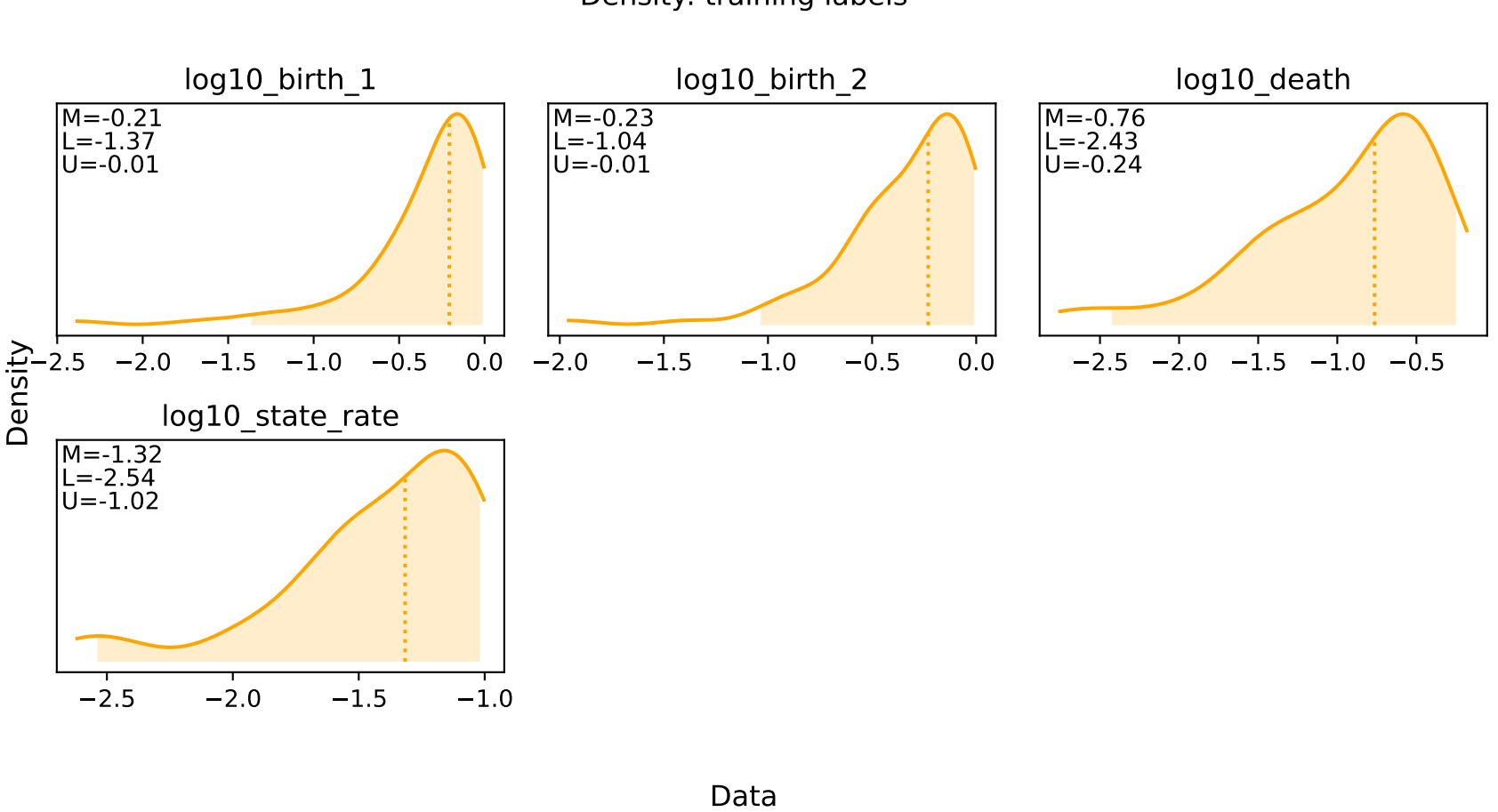
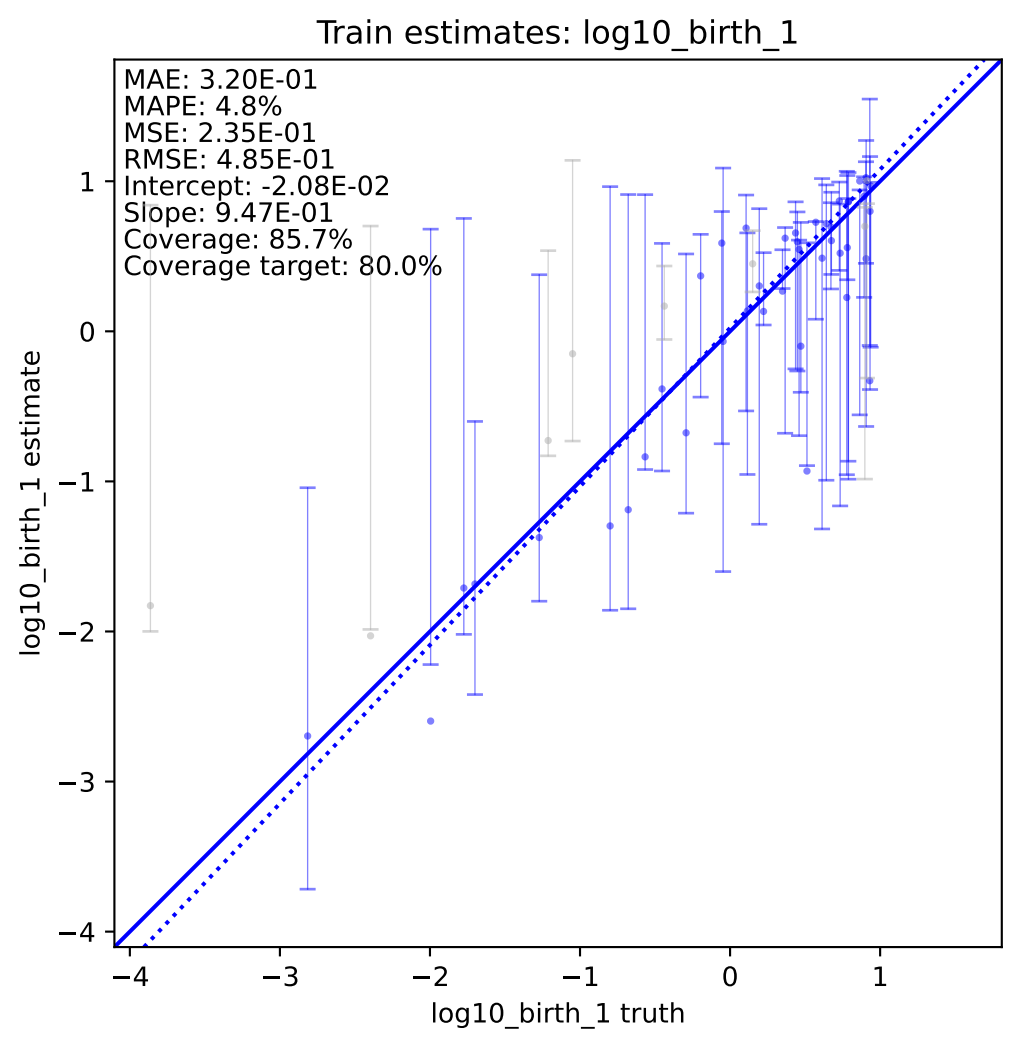
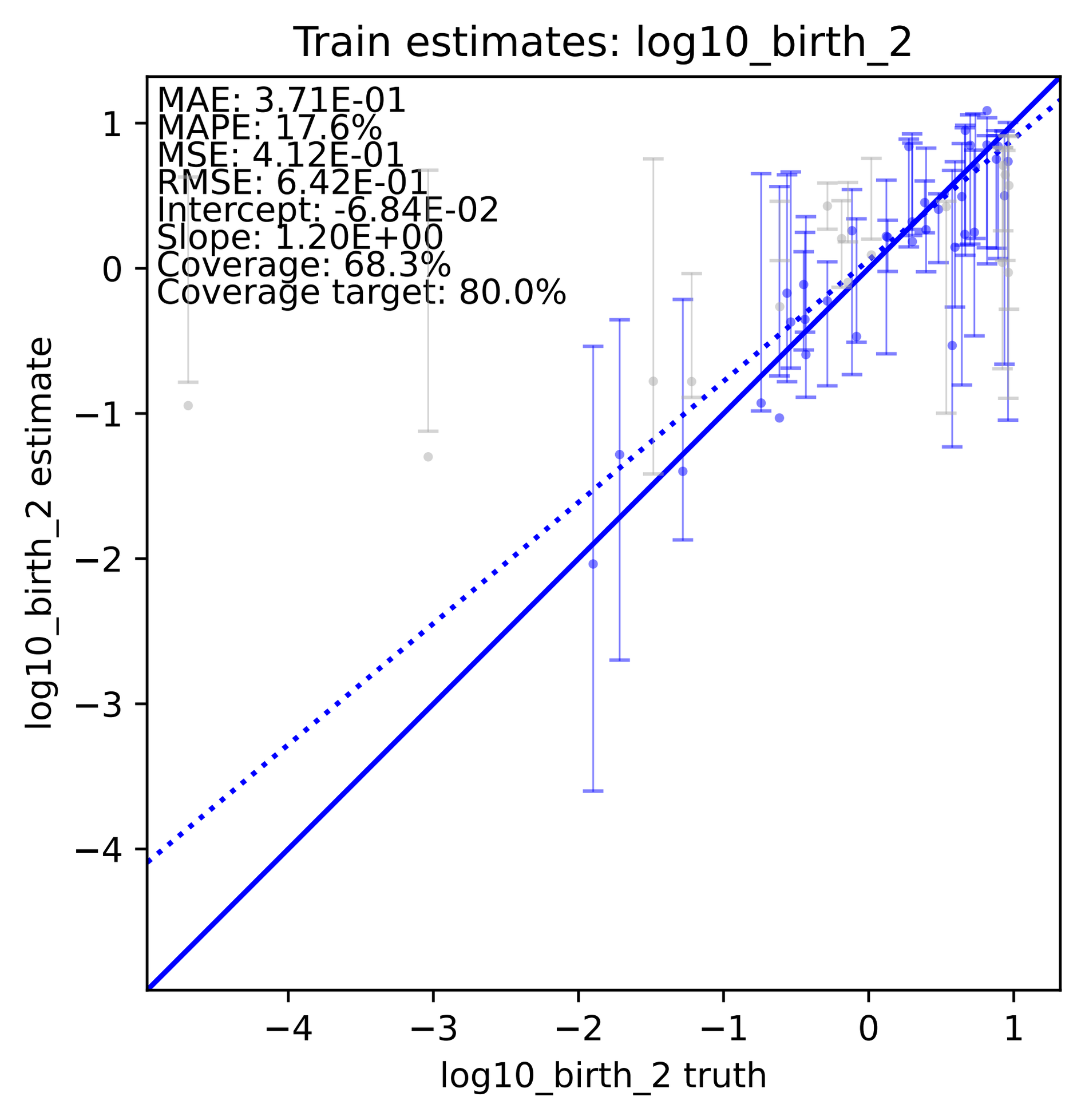

PCA: training aux. data
PC2 (25%)
PC3 (12%)
PC4 (9%)
PC1 (43%)
PC2 (25%)
PC3 (12%)

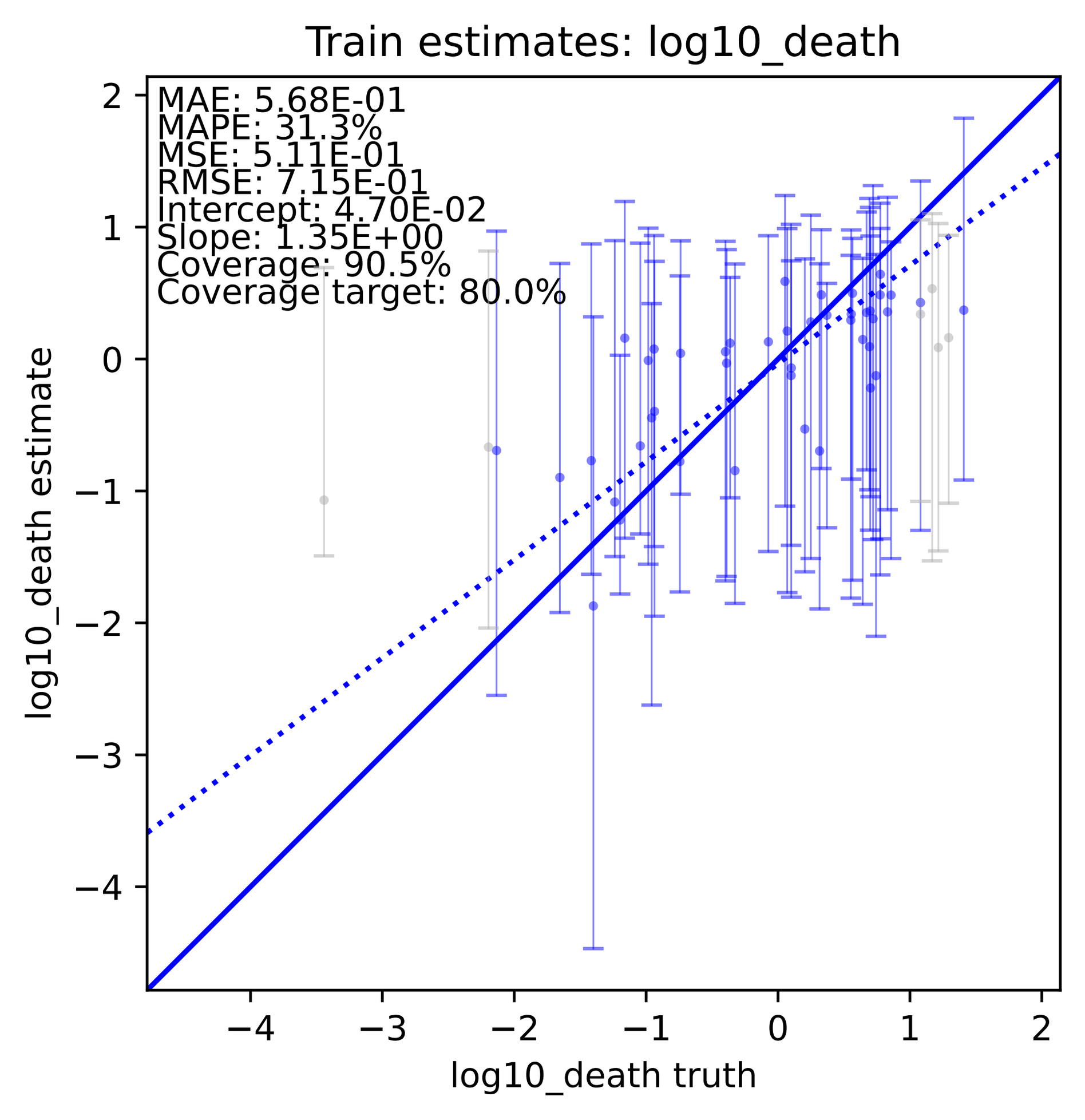

PCA: training labels
PC2 (25%)
PC3 (18%)
PC4 (9%)
PC1 (46%)
PC2 (25%)
PC3 (18%)

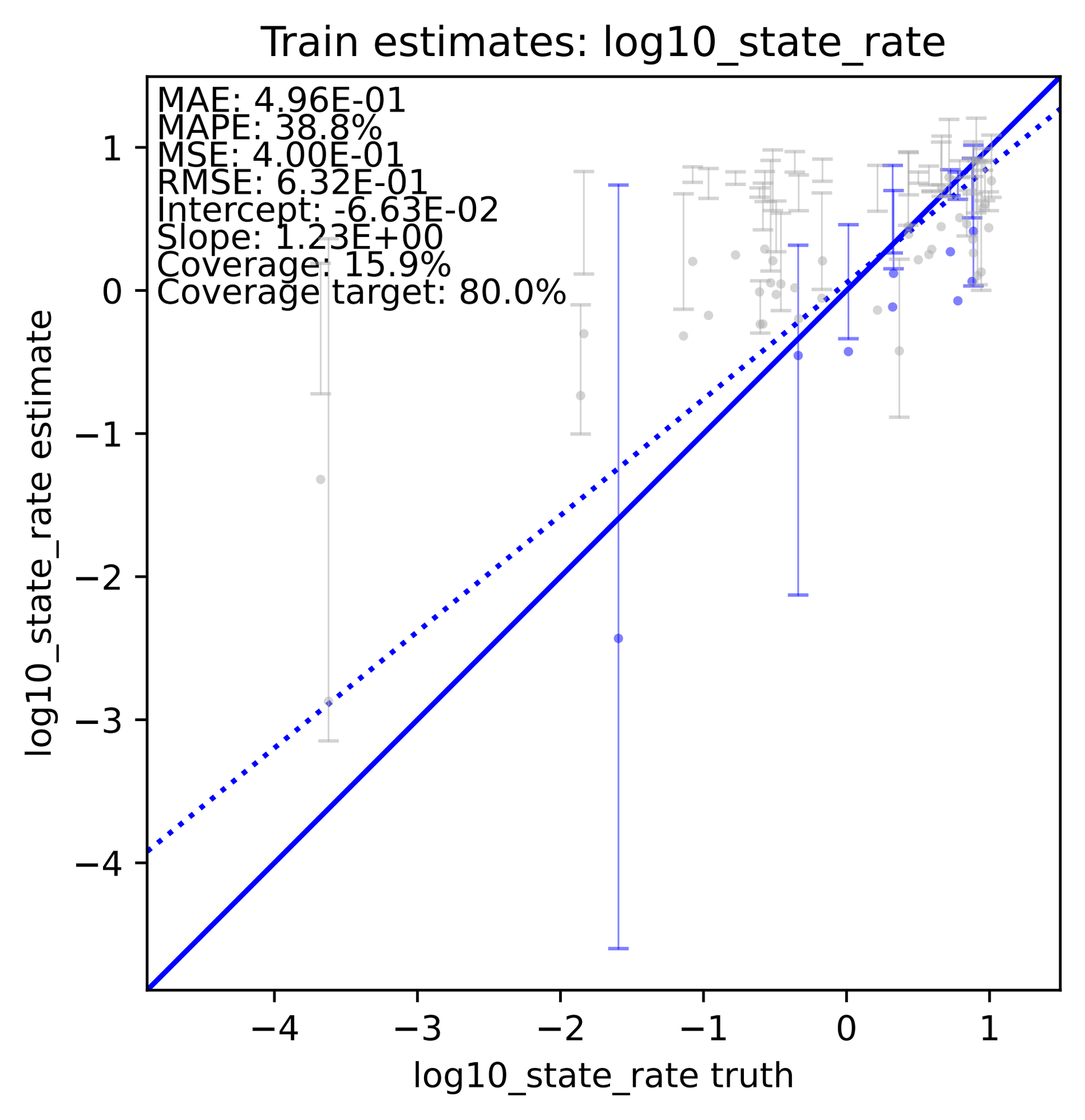

Density: training aux. data
B1
M=263.86
L=6.59
U=271.95
X=263.34
Q=44%
N_bar
M=11.39
L=4.79
U=13.83
X=11.45
Q=51%
age_mean
M=3.85
L=1.19
U=15.47
X=3.86
Q=50%
age_var
M=8.24
L=1.47
U=159.40
X=4.96
Q=30%
brlen_mean
M=1.94
L=0.63
U=8.11
X=1.94
Q=50%
brlen_var
M=2.90
L=0.51
U=48.39
X=2.33
Q=36%
colless
M=0.02
L=0.02
U=0.34
X=0.02
Q=27%
log10_sample_frac
M=-0.70
L=-0.99
U=0.00
X=-0.95
Q=11%
n_dat_0
M=196.00
L=8.00
U=476.00
X=316.00
Q=74%
n_dat_1
M=188.00
L=1.00
U=451.00
X=184.00
Q=48%
num_taxa
M=500.00
L=14.00
U=500.00
X=500.00
Q=28%
prop_taxa
M=1.00
L=1.00
U=1.00
X=1.00
Q=47%
root_age
M=15.85
L=5.57
U=58.64
X=10.85
Q=23%
tree_length
M=1575.92
L=81.25
U=5108.97
X=1935.78
Q=64%
treeness
M=0.40
L=0.29
U=0.57
X=0.33
Q=8%
Density
Data

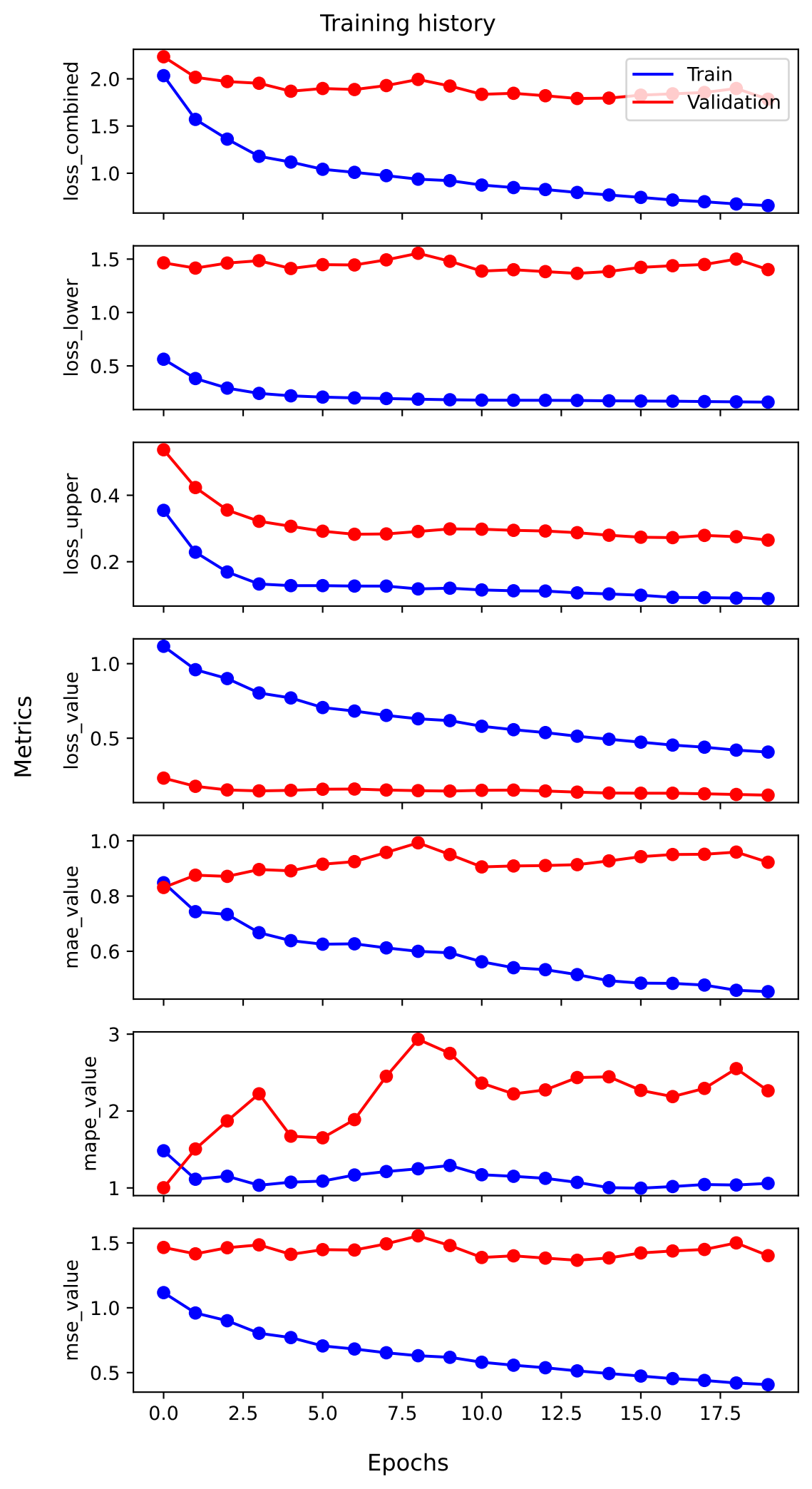

Density: training labels
log10_birth_1
M=-0.21
L=-1.37
U=-0.01
log10_birth_2
M=-0.23
L=-1.04
U=-0.01
log10_death
M=-0.76
L=-2.43
U=-0.24
log10_state_rate
M=-1.32
L=-2.54
U=-1.02
Density
Data

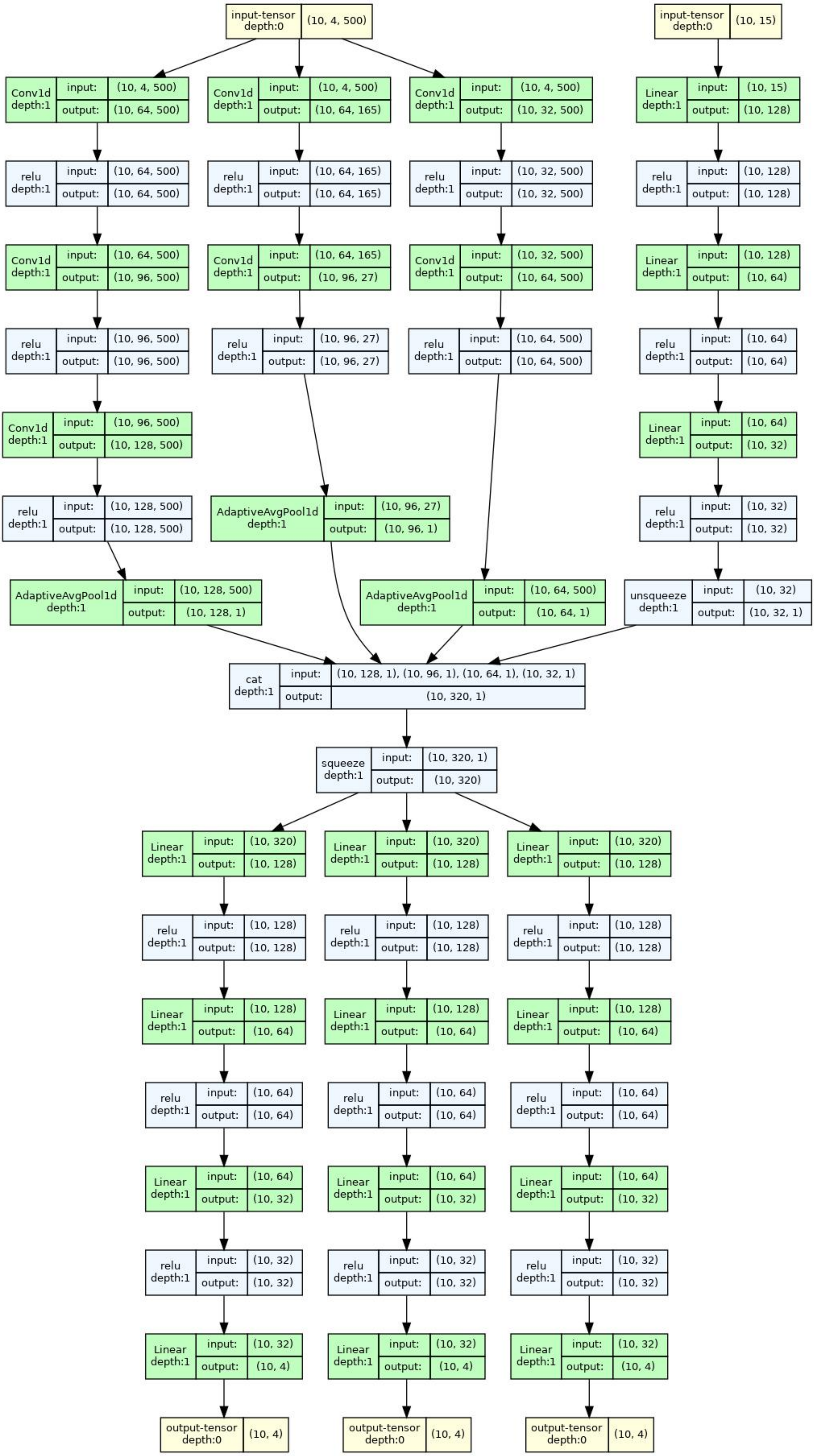

Train estimates: log10_birth_1
MAE: 3.20E-01
MAPE: 4.8%
MSE: 2.35E-01
RMSE: 4.85E-01
Intercept: -2.08E-02
Slope: 9.47E-01
Coverage: 85.7%
Coverage target: 80.0%
log10_birth_1 estimate
log10_birth_1 truth
1
0
−1
−2
−3
−4
−4
−3
−2
−1
0
1


Train estimates: log10_birth_2
MAE: 3.71E-01
MAPE: 17.6%
MSE: 4.12E-01
RMSE: 6.42E-01
Intercept: -6.84E-02
Slope: 1.20E+00
Coverage: 68.3%
Coverage target: 80.0%
log10_birth_2 estimate
log10_birth_2 truth
1
0
−1
−2
−3
−4
−4
−3
−2
−1
0
1


Train estimates: log10_death
MAE: 5.68E-01
MAPE: 31.3%
MSE: 5.11E-01
RMSE: 7.15E-01
Intercept: 4.70E-02
Slope: 1.35E+00
Coverage: 90.5%
Coverage target: 80.0%
log10_death estimate
log10_death truth
2
1
0
−1
−2
−3
−4
−4
−3
−2
−1
0
1
2


Train estimates: log10_state_rate
MAE: 4.96E-01
MAPE: 38.8%
MSE: 4.00E-01
RMSE: 6.32E-01
Intercept: -6.63E-02
Slope: 1.23E+00
Coverage: 15.9%
Coverage target: 80.0%
log10_state_rate estimate
log10_state_rate truth
1
0
−1
−2
−3
−4
−4
−3
−2
−1
0
1


Training history
Train
Validation
loss_combined
loss_lower
loss_upper
loss_value
mae_value
mape_value
mse_value
Metrics
Epochs
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


input-tensor depth:0 (10, 4, 500)
input-tensor depth:0 (10, 15)
Conv1d depth:1 input: (10, 4, 500) output: (10, 64, 500)
Conv1d depth:1 input: (10, 4, 500) output: (10, 64, 165)
Conv1d depth:1 input: (10, 4, 500) output: (10, 32, 500)
Linear depth:1 input: (10, 15) output: (10, 128)
relu depth:1 input: (10, 64, 500) output: (10, 64, 500)
relu depth:1 input: (10, 64, 165) output: (10, 64, 165)
relu depth:1 input: (10, 32, 500) output: (10, 32, 500)
relu depth:1 input: (10, 128) output: (10, 128)
Conv1d depth:1 input: (10, 64, 500) output: (10, 96, 500)
Conv1d depth:1 input: (10, 64, 165) output: (10, 96, 27)
Conv1d depth:1 input: (10, 32, 500) output: (10, 64, 500)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 96, 500) output: (10, 96, 500)
relu depth:1 input: (10, 96, 27) output: (10, 96, 27)
relu depth:1 input: (10, 64, 500) output: (10, 64, 500)
relu depth:1 input: (10, 64) output: (10, 64)
Conv1d depth:1 input: (10, 96, 500) output: (10, 128, 500)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 128, 500) output: (10, 128, 500)
AdaptiveAvgPool1d depth:1 input: (10, 96, 27) output: (10, 96, 1)
relu depth:1 input: (10, 32) output: (10, 32)
AdaptiveAvgPool1d depth:1 input: (10, 128, 500) output: (10, 128, 1)
AdaptiveAvgPool1d depth:1 input: (10, 64, 500) output: (10, 64, 1)
unsqueeze depth:1 input: (10, 32) output: (10, 32, 1)
cat depth:1 input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1) output: (10, 320, 1)
squeeze depth:1 input: (10, 320, 1) output: (10, 320)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
Linear depth:1 input: (10, 32) output: (10, 4)
Linear depth:1 input: (10, 32) output: (10, 4)
Linear depth:1 input: (10, 32) output: (10, 4)
output-tensor depth:0 (10, 4)
output-tensor depth:0 (10, 4)
output-tensor depth:0 (10, 4)